

Clinical Trials Phase Transition

A Survival-Analytic Approach on Public Registry Data

Goal

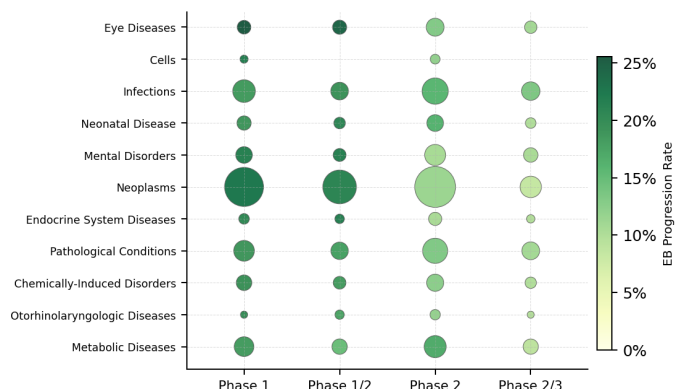
Given a program (drug X for disease Y at Phase Z), what is our best estimate of its probability of advancing to the next phase, and how confident are we? Which programs stand out as high-confidence bets, high-risk/high-reward bets, or emerging signals worth early attention?

Methodology

1. Resolved 583k raw CTGov trial records to canonical drug (RxNorm) and disease (MeSH) identities, aggregating fragmented sponsor-entered names into 51.4k clean drug-disease-phase programs
2. Modeled phase progression as a right-censored survival process, since most programs have not yet reached a resolvable outcome, avoiding the bias of penalizing recently-started programs as failures
3. Fit Kaplan-Meier curves per phase and per phase-disease group to derive a continuous, duration-aware progression probability for every program, extracting partial information from both completed and still-ongoing trials
4. Applied Empirical Bayes shrinkage (Beta-Binomial, phase-specific priors) to control small-sample noise, preventing thinly-evidenced programs from dominating rankings just for one lucky trial
5. Classified programs into five strategic profiles (high confidence, high risk high reward, emerging signals, stable but most, standard) by jointly evaluating rate, evidence volume, and estimate precision, turning a single black-box score into an interpretable, risk-stratified decision tool

Key Insights

- The dataset is dominated by Neoplasms programs (8.3k in Ph.1, 4.6k in Ph.1/2, 10k in Ph.2)
- The Eye Diseases area shows the highest EB rate (25.5% in Ph.1, 24.4% in Ph.1/2) but supported by a relatively low number of programs (126, 145) suggesting promising unexplored opportunities
- The high confidence programs are established, extensively studied oncology therapeutic, e.g. gemcitabine in pancreatic cancer
- The emerging signal programs show promising rates on very little evidence, small enough that the number could move significantly with a few more trials. They are candidates for a modest early bet, to test whether the signal holds, e.g. pazopanib in macular degeneration
- The high risk high reward programs are candidates for staged investment: e.g. latanoprost in glaucoma is worth funding for additional trials to de-risk before a larger licensing commitment
- The stable but modest and standard programs are reasonably evidenced but unremarkable, e.g. ibuprofen for pain



18.07.2026, Dr. V. Ganci, [GitHub Repo](#)

Drug	MeSH	Phase	EB rate	CI_{low}	CI_{up}	Trials	Strategic profile
pazopanib	macular degeneration	Ph.1	0.21	0.10	0.34	3	emerging signal
gemcitabine	pancreatic neoplasms	Ph.1	0.27	0.19	0.36	76	high confidence
latanoprost	glaucoma	Ph.1/2	0.21	0.10	0.35	4	high risk high reward
d-glucose	obesity	Ph.1	0.20	0.09	0.33	4	stable but modest
obinutuzumab	lymphoma, large b-cell	Ph.1	0.20	0.09	0.33	3	standard